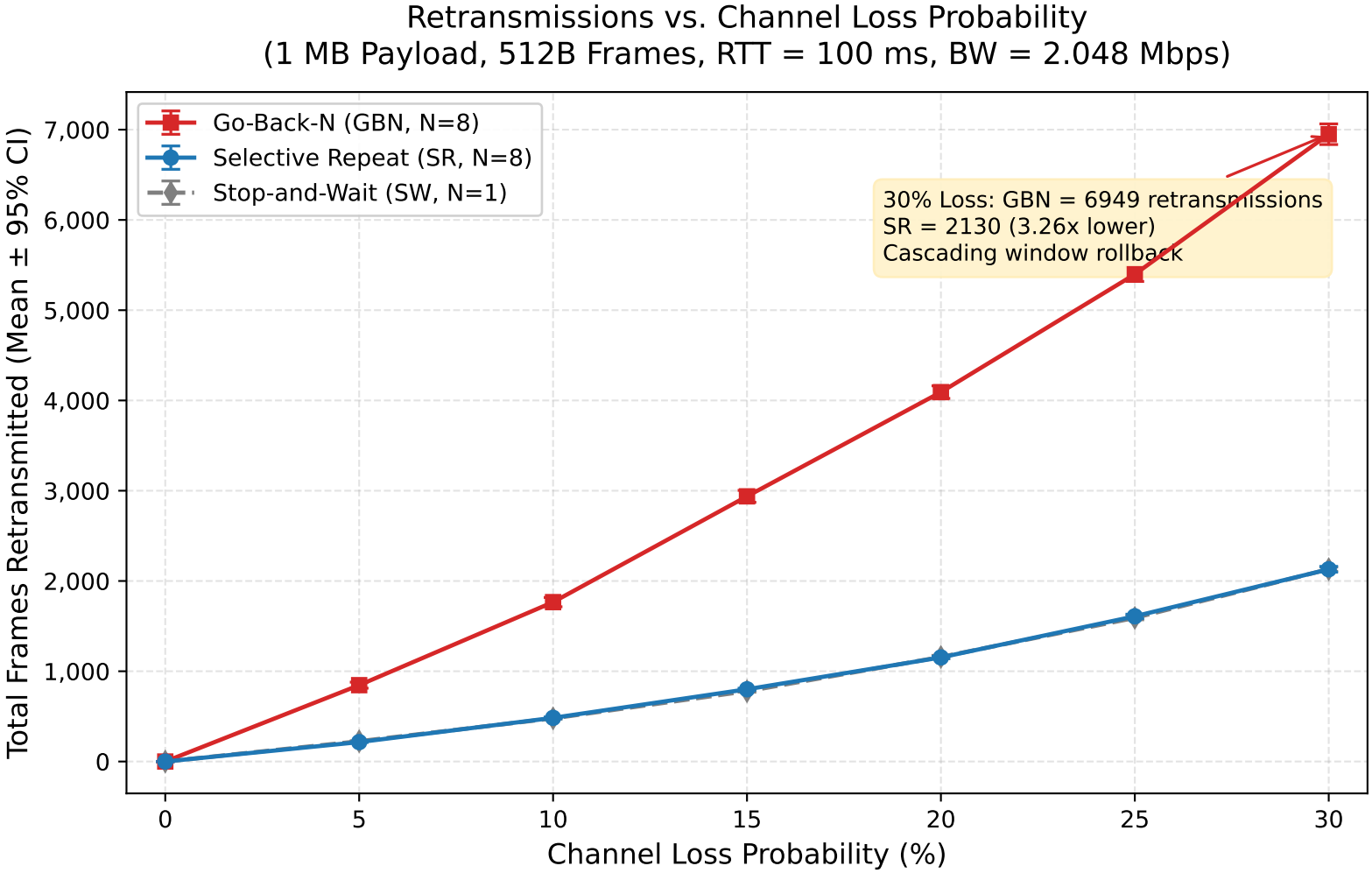
Retransmissions vs. Channel Loss Probability
(1 MB Payload, 512B Frames, RTT = 100 ms, BW = 2.048 Mbps)
Go-Back-N (GBN, N=8)
Selective Repeat (SR, N=8)
Stop-and-Wait (SW, N=1)
30% Loss: GBN = 6949 retransmissions
SR = 2130 (3.26x lower)
Cascading window rollback
Total Frames Retransmitted (Mean ± 95% CI)
Channel Loss Probability (%)
0
1,000
2,000
3,000
4,000
5,000
6,000
7,000
0
5
10
15
20
25
30